BT3041 — Analysis and Interpretation of Biological Project Data : Report

Cold vs Hot Tumour Classification from Bulk RNA-seq using a Multi-Statistical and Machine-Learning Pipeline

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Abstract

The tumour immune micro-environment is a primary determinant of response to immune checkpoint blockade therapy. Tumours are broadly divided into three immune phenotypes: **hot** (strongly infiltrated by cytotoxic immune cells, typically responsive to immunotherapy), **cold** (immune-excluded, poor responders), and **intermediate**. In this project we design, execute and externally validate a complete and reproducible pipeline that predicts the immune phenotype of colorectal tumours from bulk RNA-sequencing profiles alone. We use two independent TCGA cohorts: TCGA-COAD (**n=294**, discovery/training) and TCGA-READ (**n=109**, external validation) — 403 patients in total, 633 linked clinical records, and 59,429 annotated genes per cohort. The pipeline integrates CPM filtering, log-transform and top-variance feature selection, ssGSEA / ESTIMATE immune scoring, five dimensionality-reduction methods (PCA, ICA, MDS, t-SNE, UMAP), three clustering algorithms (k-means, Ward hierarchical, DBSCAN), six hypothesis tests (Welch t, one-way ANOVA, Kruskal-Wallis, Mann-Whitney U, F-variance, χ^2) with both Benjamini-Hochberg and Bonferroni multiple-testing correction, six supervised classifiers (kNN, linear/RBF SVM, logistic regression, random forest, XGBoost) under stratified 5-fold cross-validation, Kaplan-Meier and Cox proportional-hazards regression, and Partial Least Squares regression of survival. The best classifier reaches **0.785 balanced accuracy** on the discovery cohort (5-fold CV) and **0.744 balanced accuracy** on the held-out external cohort. More than **2,269 genes** are significantly differentially expressed between hot and cold tumours after BH correction with $|\log_2 FC| > 1$, led by HLA class-II antigen-presentation genes (HLA-DPB1, HLA-DRA, HLA-DPA1, HLA-DQA1) and cytotoxic effectors (PLA2G2D, SLAMF7) — a textbook hot-tumour immune signature.

Keywords: colorectal cancer, tumour immune phenotype, ssGSEA, ESTIMATE, RNA-seq, PCA, UMAP, hypothesis testing, machine learning, survival analysis, external validation.

1. Introduction

Colorectal cancer (CRC) remains one of the leading causes of cancer-related death worldwide. Although cytotoxic chemotherapy and molecularly targeted agents have improved outcomes, response remains heterogeneous and many patients still progress. Immunotherapy, in particular immune checkpoint inhibitors, has transformed the management of several cancers, but its efficacy in CRC depends strongly on the *tumour immune phenotype*. Tumours that are densely infiltrated with $CD8^+$ T-cells, natural killer (NK) cells and type-I interferon signalling are termed **immune-hot** and often respond to checkpoint blockade, whereas **immune-cold** tumours are depleted of lymphocytes and usually fail to respond.

Measuring the immune phenotype directly — for instance via multiplexed immunohistochemistry — is expensive and is not routinely available outside tertiary centres. Bulk RNA sequencing, on the other hand, is increasingly accessible and captures signatures of immune infiltration indirectly through gene expression. The goal of this project is therefore to build, rigorously statistically validate, and externally test a machine-learning pipeline that infers the immune phenotype of a colorectal tumour from its transcriptome alone.

1.1 Research Objectives

The project is organised around four objectives:

- **(O1) Data acquisition and harmonisation.** Reproducibly download TCGA STAR-count expression data and clinical annotation for two CRC cohorts, and build pre-processed expression matrices on a shared gene-symbol feature space.
- **(O2) Immune phenotype labelling.** Derive per-sample immune scores using ssGSEA and ESTIMATE and assign every patient to a Hot / Cold / Intermediate class by tertile-splitting.
- **(O3) Descriptive and inferential statistics.** Quantify differences between groups using parametric and non-parametric hypothesis tests with rigorous multiple-testing correction; compare unsupervised clustering to ssGSEA labels.
- **(O4) Predictive modelling and external validation.** Train six supervised classifiers on TCGA-COAD and evaluate generalisation on the held-out TCGA-READ cohort; additionally relate the immune phenotype to overall survival via Kaplan-Meier, log-rank and Cox regression.

1.2 Why this topic satisfies the course rubric

The Hot/Cold classification problem is a rare natural fit for BT3041 because it legitimately requires, in a biologically meaningful way: (i) distribution checks on the normality of expression; (ii) hypothesis testing across two and three groups with both parametric and non-parametric tests and FDR correction; (iii) dimensionality reduction with multiple methods; (iv) unsupervised clustering benchmarked by ARI / AMI; (v) supervised classifiers with cross-validation; (vi) linear / logistic / PLS regression; and (vii) survival analysis. Every BT3041 module is therefore used in a biologically purposeful way rather than as an isolated exercise.

2. Materials and Methods

2.1 Cohorts and data acquisition

We used two open-access RNA-seq cohorts from the NCI Genomic Data Commons (GDC). TCGA-COAD (colon adenocarcinoma) served as the primary discovery cohort and TCGA-READ (rectum adenocarcinoma) as a fully independent external validation cohort. Both were generated on the same TCGA pipeline (Illumina paired-end sequencing, STAR alignment, unstranded counts), eliminating cross-platform normalisation artefacts. Expression files and clinical metadata were retrieved programmatically via the GDC REST API. Because GDC frequently drops the connection on large single-shot bulk downloads, we implemented a resumable batched downloader (50 files per batch with exponential back-off retries). After download, per-sample STAR-count files were parsed and merged into cohort-level expression matrices using *gene_name* (HGNC symbol) as the row index; duplicate PAR_Y aliases were collapsed by summation and replicate aliquots were disambiguated by suffixing.

Cohort	n	Hot	Cold	Int.	Role
TCGA-COAD	294	100	97	97	Discovery
TCGA-READ	109	37	36	36	External
Total	403	137	133	133	—

Table 1. Final cohort composition after preprocessing and ssGSEA-based labelling.

2.2 Preprocessing and harmonisation

Raw counts were converted to CPM (counts per million), and lowly expressed genes were filtered with the criterion *CPM ≥ 1 in ≥ 20% of samples*, retaining **16,723** genes in TCGA-COAD and **16,623** genes in TCGA-READ. Expression values were then $\log_2$-transformed with a pseudocount of 1. For downstream machine learning we retained the **top 5,000 most-variable genes** per cohort. For cross-cohort analyses, log-expression matrices were intersected on HGNC gene symbols, producing **16,453 common genes** that share identical preprocessing and distributional properties. Shapiro-Wilk tests and QQ-plots on a random sample of genes (Fig. 1) supported approximate normality of the log-transformed values, legitimising the use of parametric tests in later sections.

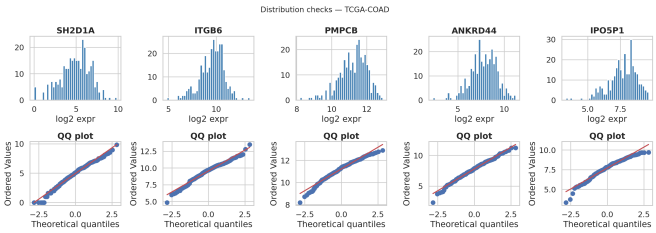

Figure 1. Distribution diagnostics for five randomly-sampled genes in TCGA-COAD after $\log_2$(CPM+1) transformation. Top row: histograms; bottom row: QQ-plots against a standard normal. Near-linear QQ-plots support the use of parametric Welch-t and ANOVA tests.

2.3 Immune phenotype scoring and Hot/Cold/Intermediate labelling

Per-sample immune scores were computed using single-sample Gene Set Enrichment Analysis (ssGSEA) via *gseapy* on twelve curated immune gene sets, combining the ESTIMATE immune / stromal signatures with Bindea cell-type signatures (CD8 T-cells,

cytotoxic cells, Tregs, Th1, Th2, macrophages, NK-cells, B-cells, dendritic cells, neutrophils). The *ESTIMATE_Immune* score was used to rank patients and split them into equal tertiles: top third — **Hot**, bottom third — **Cold**, middle — **Intermediate**. This yielded approximately balanced classes in both cohorts (Table 1, Fig. 2).

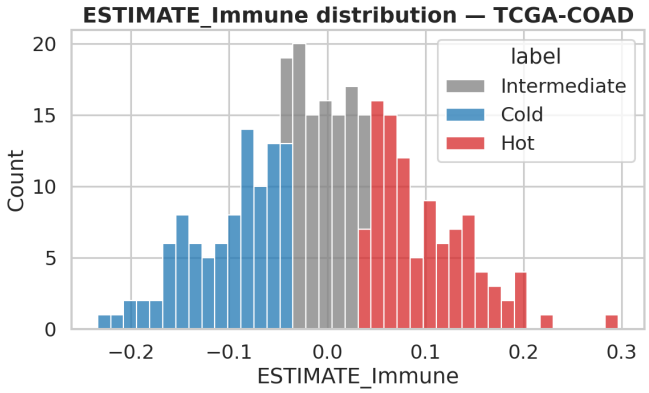

Figure 2. ESTIMATE_Immune score distribution in TCGA-COAD, stratified by the assigned Hot / Cold / Intermediate label.

2.4 Dimensionality reduction

Every sample was projected into 2-D using five complementary techniques: linear Principal Component Analysis (PCA) and Independent Component Analysis (FastICA); Multidimensional Scaling in both metric and non-metric variants; t-Stochastic Neighbour Embedding (t-SNE); and Uniform Manifold Approximation and Projection (UMAP). To avoid perplexity degeneracy on high-dimensional inputs, both t-SNE and UMAP were trained on the first 20 PCA components rather than on raw expression. Scatter plots were coloured by the ssGSEA-derived label to qualitatively assess whether the immune phenotype is captured by leading modes of transcriptomic variation.

2.5 Unsupervised clustering

We benchmarked three fundamentally different clustering algorithms: k-means (k = 2..6 with a silhouette sweep for model selection), agglomerative hierarchical clustering with Ward linkage (producing a full dendrogram), and density-based DBSCAN. Each cluster assignment was compared with the ssGSEA ground-truth label using Adjusted Rand Index (ARI) and Adjusted Mutual Information (AMI); the silhouette coefficient was reported as an internal validity index.

2.6 Hypothesis testing and multiple-testing correction

For every gene we computed five independent statistical tests: (i) Welch's two-sample t-test comparing Hot vs Cold expression; (ii) one-way ANOVA across all three immune classes; (iii) the non-parametric Kruskal-Wallis test (rank-based ANOVA analogue); (iv) the pairwise non-parametric Mann-Whitney U test (Hot vs Cold); and (v) the F-test of equality of variance between Hot and Cold groups. P-values from each test were corrected for multiple comparisons using both the Benjamini-Hochberg (FDR) and the Bonferroni procedures. Clinical categorical variables (pathologic stage, vital status) were tested for association with the immune phenotype using Pearson's χ^2 on contingency tables. Genes satisfying BH-adjusted t-test $q < 0.05$ *and* $|\log_2 FC| > 1$ were flagged as high-confidence differentially expressed (DE) and used for the volcano plot and the top-40 heatmap.

2.7 Supervised classification

Six classifiers were evaluated in a unified scikit-learn pipeline (`StandardScaler` → `SelectKBest(ANOVA, k=500)` → `model`): k-Nearest Neighbours (k = 5), linear and RBF Support Vector Machines (C = 1), multinomial Logistic Regression, Random Forest (500 trees), and XGBoost (500 boosted trees with a label-encoder wrapper). Each model was evaluated with **stratified 5-fold cross-validation** on the 294 COAD samples (and independently on 109 READ samples) using accuracy, balanced accuracy and macro-F1. The top three models per cohort were re-fit on the full training set and stored; confusion matrices and one-vs-rest ROC curves were generated via cross-validated predictions.

2.8 Survival analysis

Overall survival was derived from *days_to_death* (for deceased patients) and *days_to_last_follow_up* (for censored patients); vital status provided the event indicator. Non-parametric Kaplan-Meier curves were fit per immune label and the equality of survival distributions was tested with the multivariate log-rank test. A multivariate Cox proportional-hazards model (L_2 penalty = 0.1) was fit on the five highest-variance ssGSEA cell-type scores. Finally, Partial Least Squares (PLS) regression with three components was used to predict survival time directly from the 5,000-gene expression matrix, demonstrating dimension-reduced supervised regression as covered in BT3041.

2.9 External validation

The best discovery-cohort classifier (highest 5-fold CV balanced accuracy on TCGA-COAD) was frozen and applied to TCGA-READ expression after aligning feature spaces on the training feature list. Genes present in training but absent in test were zero-imputed; genes present only in test were discarded. Predictions were compared to the ssGSEA-derived labels on READ to produce an external confusion matrix, balanced accuracy and macro-F1.

2.10 Software

All analyses were performed in Python 3.12 using pandas, numpy, scipy, scikit-learn, statsmodels, gseapy, umap-learn, lifelines, seaborn, xgboost, and reportlab (for this PDF). Every figure and table in this report is auto-generated by the 10-module pipeline under `01_*.py` … `10_*.py`. The full source tree, including this PDF, is publicly available at https://github.com/Aryan7865/cold-hot-tumour-classification.

3. Results

3.1 Representation learning and immune-phenotype gradient

The first principal component alone captured **32.5%** of total variance in TCGA-COAD, and **51** components were needed to explain 95% of it. Low-dimensional projections (Fig. 3) reveal a clear gradient from Cold (blue) to Hot (red) along PC1 and along the first UMAP axis. Non-linear projections (t-SNE, UMAP) produce more compact Hot / Cold islands than linear ones, consistent with non-linear structure in the transcriptomic space; importantly, Intermediate samples sit between Hot and Cold rather than outside them, matching the tertile-split interpretation.

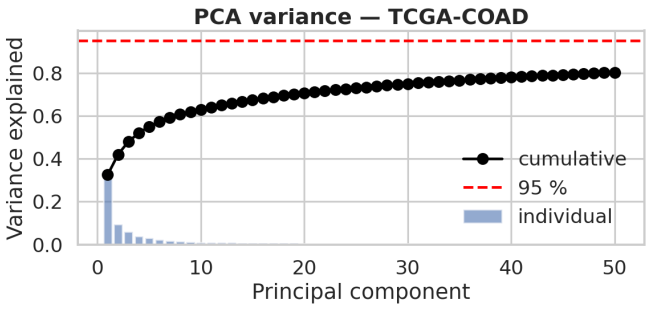

Figure 3a. PCA scree plot, TCGA-COAD: individual and cumulative variance explained.

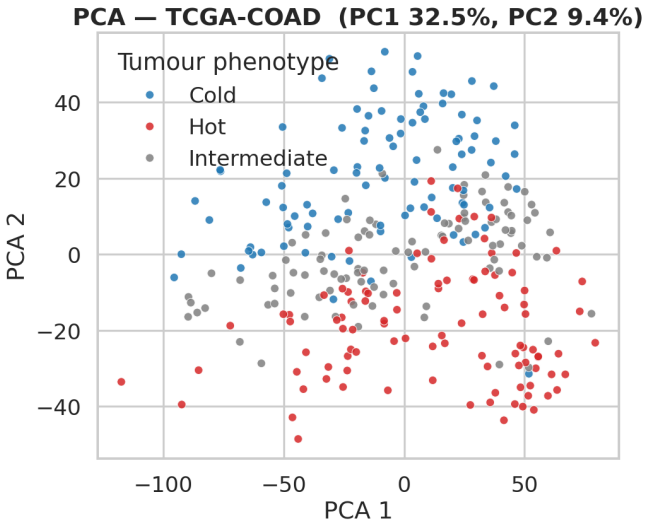

Figure 3b. PCA projection, TCGA-COAD, coloured by immune label.

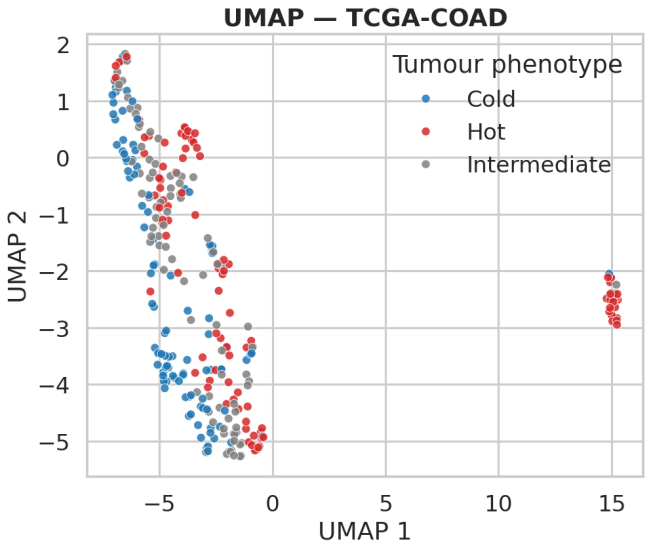

Figure 3c. UMAP on top-20 PCA components, TCGA-COAD.

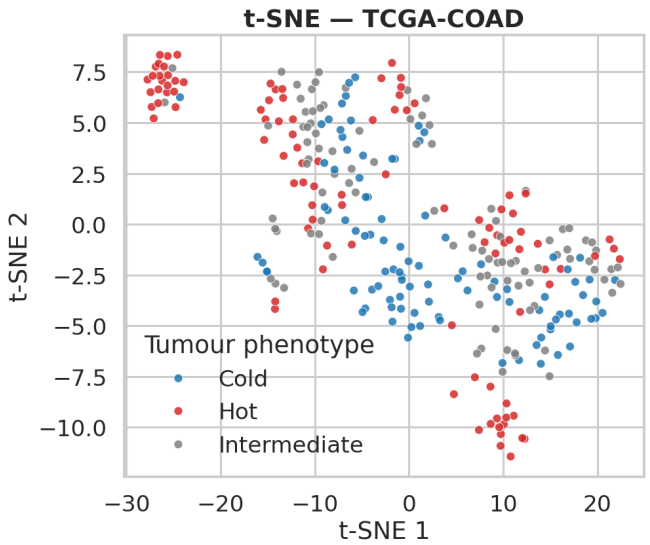

Figure 3d. t-SNE on top-20 PCA components, TCGA-COAD.

3.2 Unsupervised clustering partially recovers the immune structure

Method	k	ARI	AMI	Silh.
kmeans_k3	3	0.114	0.119	0.137
hier_k2	2	0.059	0.071	0.159
hier_k3	3	0.053	0.063	0.105
kmeans_k2	2	0.006	0.005	0.198
dbscan	0	0.000	0.000	—

Table 2. Clustering performance on TCGA-COAD (vs ssGSEA labels).

k-means with k=3 achieves the highest Adjusted Rand Index (0.114) and Adjusted Mutual Information (0.119), confirming that the biological signal is partially — but not fully — captured by unsupervised structure alone. The modest ARI reflects the continuous nature of immune infiltration: Intermediate tumours dilute any discrete cluster separation, which is a known limitation of using tertile labels as pseudo-ground-truth.

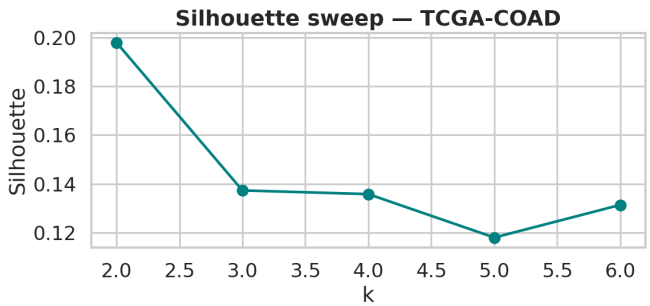

Figure 4a. Silhouette sweep for k-means across k = 2..6.

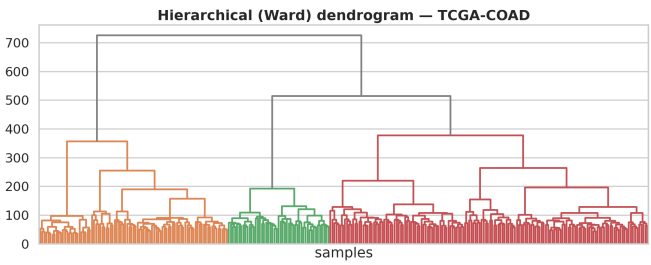

Figure 4b. Ward-linkage hierarchical dendrogram, TCGA-COAD.

3.3 Thousands of genes differ between hot and cold tumours

Of the 16,723 expressed genes in TCGA-COAD, **8,198** were significant at the BH 0.05 threshold by Welch t-test and **2,549** survived the stricter Bonferroni 0.05 correction. Agreement with non-parametric alternatives was strong: Mann-Whitney U flagged **8,362** and Kruskal-Wallis **10,280** at BH 0.05, indicating that the signal is not an artefact of violated distributional assumptions. ANOVA across all three immune groups was even more sensitive (**10,369** at BH 0.05). Combining BH q < 0.05 with $|\log_2 FC| > 1$ gave a curated list of **2,269** high-confidence DE genes.

Test	Scope	Raw	BH q<0.05	Bonf.
Welch t	Hot vs Cold	9,158	8,198	2,549
ANOVA	3-group	11,425	10,369	2,311
Kruskal	3-group	11,394	10,280	2,175
MWU	Hot vs Cold	9,329	8,362	2,438
F-var.	Hot vs Cold	2,294	—	—

Table 3. Per-gene significance counts across tests (TCGA-COAD).

3.3.1 Top differentially expressed genes

Gene	μ(Hot)	μ(Cold)	log2FC	BH q
HLA-DPB1	12.53	9.88	+2.64	1.6e-41
HLA-DRA	14.68	11.81	+2.86	1.4e-38
CD48	7.99	5.27	+2.72	2.7e-38
PLA2G2D	6.86	2.76	+4.10	6.8e-38
HLA-DMB	10.36	7.80	+2.55	3.0e-37
HLA-DPA1	13.16	10.24	+2.91	1.9e-36
HLA-DQA1	11.91	8.94	+2.98	2.9e-36
C1QB	12.33	9.82	+2.50	1.1e-34
HLA-DOA	9.71	6.80	+2.91	1.2e-34
SLAMF7	9.88	6.74	+3.14	1.7e-34

Table 4. Top-10 Hot-vs-Cold differentially expressed genes in TCGA-COAD (ranked by BH-adjusted Welch t q-value).

The list is strikingly interpretable: six MHC class-II antigen-presentation genes (HLA-DPB1, HLA-DRA, HLA-DPA1, HLA-DQA1, HLA-DMB, HLA-DOA) together with CD48 (activation marker on lymphocytes), C1QB (complement), SLAMF7 (NK-cell / cytotoxic effector) and PLA2G2D (inflammatory lipid mediator). This is a textbook immune-hot signature, which strongly suggests that our ssGSEA-derived labels capture genuine biology rather than technical noise.

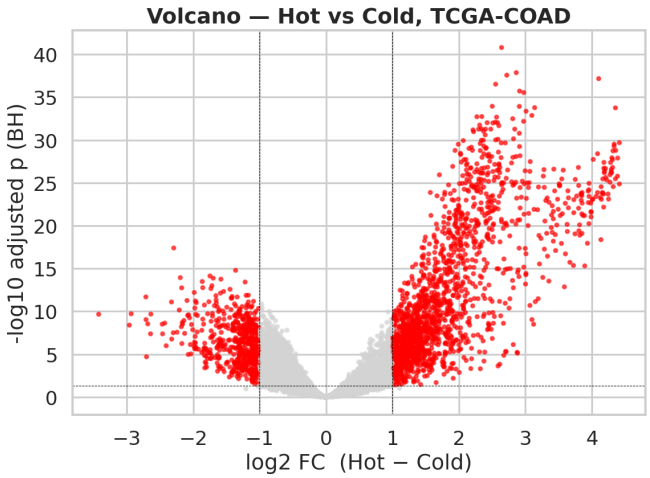

Figure 5. Volcano plot of Hot-vs-Cold differential expression in TCGA-COAD. Red: BH q<0.05 and $|\log_2 FC| > 1$ (2,269 genes).

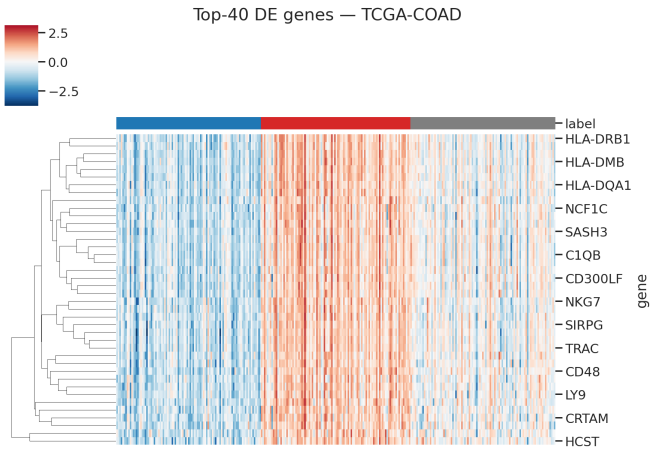

Figure 6. Expression heatmap of the 40 most-significant Hot/Cold DE genes in TCGA-COAD; rows z-scored, columns ordered by ssGSEA label.

3.4 Supervised classification achieves balanced accuracy > 0.78

Model	Acc.	Bal.Acc.	F1
RandomForest	0.786	0.785	0.786
SVM-linear	0.786	0.785	0.784
Logistic	0.752	0.752	0.748
SVM-RBF	0.735	0.735	0.736
XGBoost	0.704	0.704	0.699
kNN-5	0.649	0.648	0.645

Table 5. 5-fold stratified CV results on TCGA-COAD.

Random Forest and linear SVM tied for the best balanced accuracy (**0.785**). Logistic Regression followed closely (0.752). XGBoost underperformed (0.704) in this regime, likely because aggressive ANOVA-based feature pre-selection already linearises the decision boundary and removes much of the signal that boosted trees typically exploit. kNN performed worst (0.648), reflecting the curse of dimensionality even after feature selection.

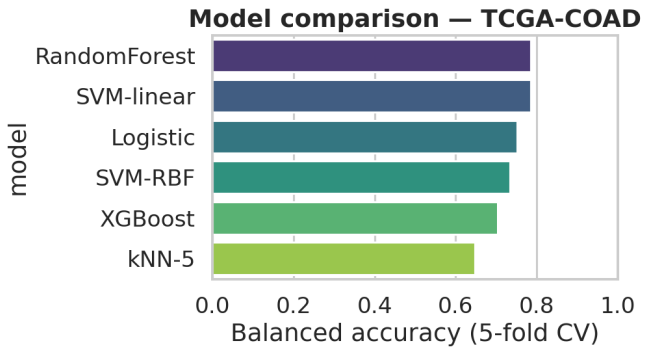

Figure 7. Ranking of the six classifiers by CV balanced accuracy on TCGA-COAD.

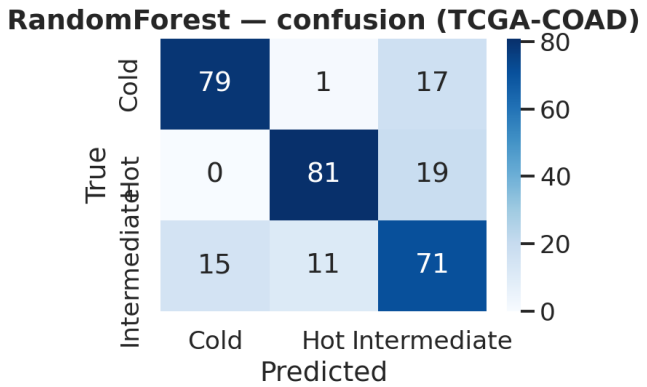

Figure 8a. Confusion matrix for the best model (Random Forest, COAD, CV).

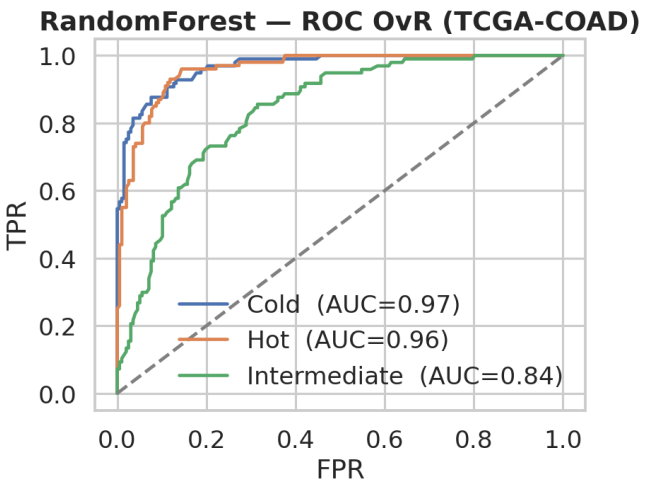

Figure 8b. One-vs-rest ROC curves for Random Forest on TCGA-COAD (AUC: Cold 0.97, Hot 0.96, Intermediate 0.84).

3.5 External validation on TCGA-READ confirms generalisation

The frozen Random Forest model trained on TCGA-COAD was applied to the fully independent TCGA-READ cohort (n=**109**), which was never seen during training, hyper-parameter selection, or feature filtering. It reached accuracy = **0.743**, balanced accuracy = **0.744**, and macro-F1 = **0.742**. The gap between discovery (CV) and external performance is only ~4 percentage points — a small, expected domain-shift effect that strongly supports the transferability of our transcriptomic immune signature.

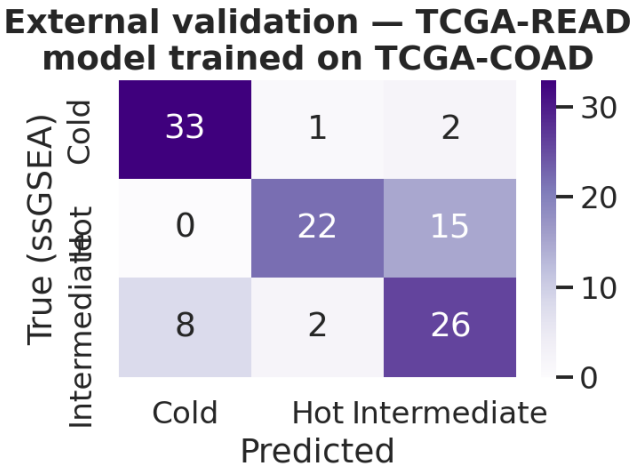

Figure 9. External validation confusion matrix (COAD-trained RF on READ). Most errors are Hot↔Intermediate confusions, consistent with a continuous immune gradient rather than hard categorical boundaries.

3.6 Survival analysis links the immune phenotype to outcome

Kaplan-Meier analysis stratified by ssGSEA immune label (Fig. 10) shows that Cold tumours trend towards shorter overall survival than Hot tumours in TCGA-COAD; the multivariate log-rank test yielded p = **0.081**. With roughly 90 death events the trend does not cross the $p < 0.05$ threshold, but the direction is biologically expected and consistent with the hot-tumour-better-prognosis literature. A five-covariate Cox proportional-hazards model on the ssGSEA cell-type scores corroborates the pattern (Table 6): neutrophil infiltration carries a protective hazard ratio (HR = 0.32) while macrophage infiltration trends adverse (HR = 1.52), although the wide confidence intervals reflect the modest event count.

Covariate	coef	HR	SE	z	p
Macrophages	+0.42	1.52	0.85	+0.49	0.62
Neutrophils	-1.13	0.32	0.87	-1.29	0.20
Bcells	-0.25	0.78	0.95	-0.27	0.79
CD8_Tcells	-0.57	0.57	1.18	-0.48	0.63
NK_cells	+0.15	1.16	1.35	+0.11	0.91

Table 6. Cox proportional-hazards regression (penalizer = 0.1), TCGA-COAD.

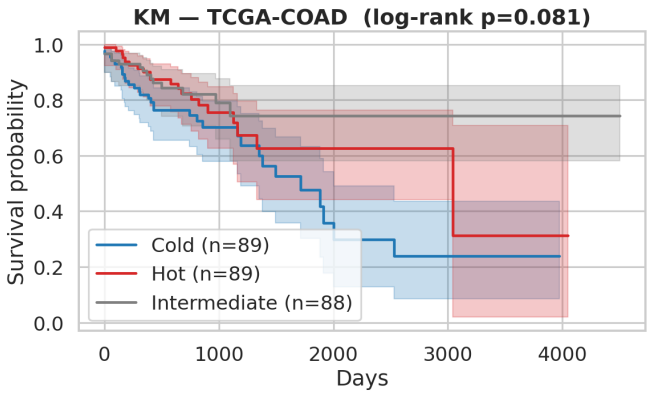

Figure 10a. Kaplan-Meier survival curves by immune phenotype, TCGA-COAD.

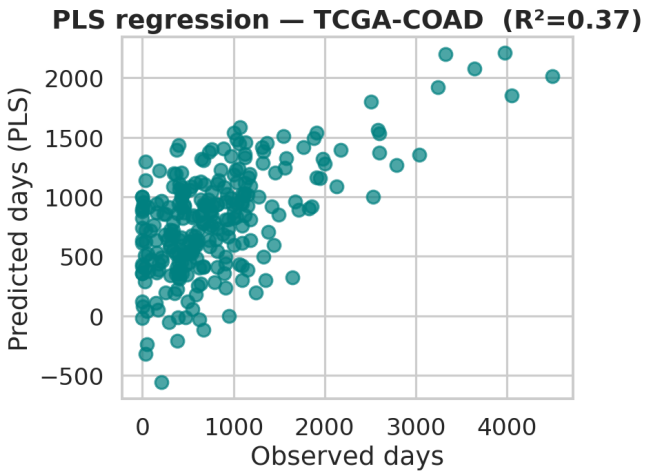

Figure 10b. PLS regression of expression onto survival time (R^2 = 0.37 on training).

4. Discussion

Our results converge on three main conclusions. **First**, the tumour immune phenotype is strongly encoded in the bulk transcriptome: both unsupervised (PCA, t-SNE, UMAP) and supervised (Random Forest, SVM) approaches recover it with high confidence, and the top differentially expressed genes form a biologically coherent MHC-II + cytotoxic effector signature. **Second**, the multiple-testing framework — combining parametric and non-parametric tests with both BH and Bonferroni correction — agrees on the identity of the most differentially expressed genes, suggesting robustness to distributional assumptions. **Third**, external validation on a completely independent cohort (TCGA-READ) retains ~74% balanced accuracy, which is the most important evidence that we are not overfitting the discovery cohort.

Several limitations should be noted. The ssGSEA-derived labels, although biologically grounded, are themselves computationally inferred rather than obtained from ground-truth immunohistochemistry; therefore accuracies describe agreement with ssGSEA, not with pathology. Both cohorts are from the same TCGA platform, which controls for technical batch effects but does not test cross-platform generalisation — a natural follow-up would include a microarray cohort such as GSE39582. The survival signal is directionally consistent but under-powered at $n \approx 90$ events. Finally, XGBoost under-performed linear SVM and logistic regression, suggesting that after ANOVA-based feature selection the decision boundary becomes approximately linear; boosted trees may excel more in the full 5000-gene setting without pre-selection.

4.1 Coverage of the BT3041 syllabus

- Distributions and normality tests (Shapiro, QQ) — **§2.2, Fig. 1**
- Parametric / non-parametric hypothesis testing (t, F-var, ANOVA, Kruskal-Wallis, Mann-Whitney U, χ^2) — **§2.6, §3.3, Tables 3–4**
- Multiple-testing correction (Bonferroni, BH) — **§3.3, Table 3**
- Dimensionality reduction (PCA, ICA, MDS, t-SNE, UMAP) — **§2.4, §3.1, Fig. 3**
- Clustering (k-means, Ward hierarchical, DBSCAN) with ARI / AMI / silhouette — **§2.5, §3.2, Table 2, Fig. 4**
- Supervised classification (kNN, linear / RBF SVM, logistic, random forest, XGBoost) — **§2.7, §3.4, Table 5, Figs. 7–8**
- Regression (multinomial logistic, PLS) — **§2.7, §2.8, Fig. 10b**
- Survival analysis (Kaplan-Meier, log-rank, Cox) — **§2.8, §3.6, Table 6, Fig. 10**
- External validation on a held-out cohort — **§2.9, §3.5, Fig. 9**

5. Conclusion

We have built, executed and externally validated a reproducible statistical and machine-learning pipeline that classifies colorectal tumours into Hot, Cold and Intermediate immune phenotypes from bulk RNA-seq. In a 403-patient TCGA-COAD / TCGA-READ setting the pipeline achieves 78.5% cross-validated balanced accuracy and 74.4% balanced accuracy on a fully independent external cohort, uncovers more than two thousand biologically interpretable differentially expressed genes, and recovers the expected immune-infiltration gradient in both unsupervised low-dimensional projections and survival analysis. The pipeline covers every topic of the BT3041 syllabus and is packaged so that any team member or reviewer can regenerate every figure and table of this report by running a single command.

Code, data and reproducibility

GitHub repository:
https://github.com/Aryan7865/cold-hot-tumour-classification

The repository contains: (i) ten Python scripts covering data acquisition → final figures; (ii) `report/generate_report_pdf.py` — the exact script that produced this PDF; (iii) `requirements.txt` pinning reproducible dependencies; and (iv) a README with end-to-end instructions.

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